



# Course Information

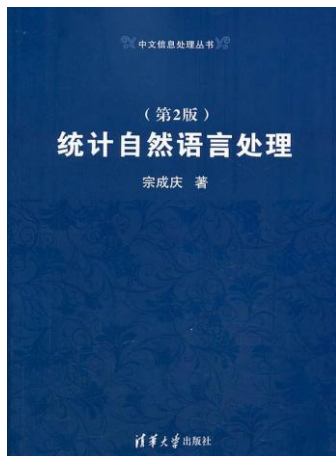
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- Lecture
  - Prof. Xu Ruifeng 徐睿峰
  - Office: L1602
  - Email: [xuruifeng@hit.edu.cn](mailto:xuruifeng@hit.edu.cn)
- Teaching Assistant
  - 郎骐玮 L1407



# Textbook

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- 宗成庆 《统计自然语言处理》  
第二版



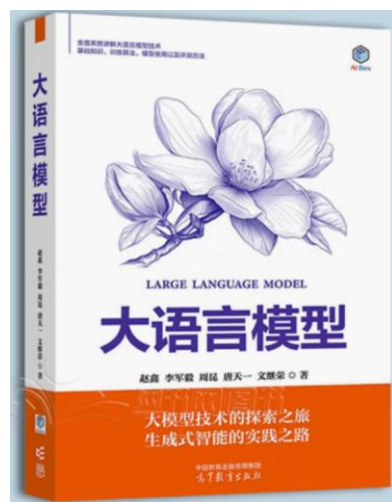
- 刘挺 秦兵 赵军等 《自然语言处理》 2021

# Textbook

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- 张奇 桂韬 黄萱菁等 《自然语言处理导论》 2023



- 赵鑫等 《大语言模型》 2025

# Course Grading

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- Grading
  - Course Project 40%
    - System Design and Implementation
    - Grading based on
      - Performance 70%
      - Report and Demonstration 10%
      - Innovative Ideas 20%
    - Collaboration among students
      - Three students in each group
  - Final Examination 60%

# Questions that today's class will answer

- What is Natural Language Processing (NLP)?
- What can NLP technique do?
- Why NLP is hard?
- Can we build system that learn from text?
- How to start NLP research
- What will this course be about?

# Natural Language

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The process that handling the *SOMETHING*

- **Natural language** is a language that is spoken, written, or signed by humans for general-purpose communication- wikipedia
- What kinds of things do people say?
- What do these things say/ask/request about the world?

# Natural Language: Multi-media

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- Text form:
  - Plain text, RTF text, Image text
- Audio form:
  - speech
- Video form, a mixed form
  - text, speech



# Natural Language : Multi-lingual

- Babel
- Bible Genesis



# Natural Language : Multi-lingual

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- We have more than **150** languages on the earth

**“I love you”** in 35 languages

|                          |                       |                                   |             |
|--------------------------|-----------------------|-----------------------------------|-------------|
| <i>I love you</i>        | (English)             |                                   |             |
| 我爱你                      | (Chinese simplified)  |                                   |             |
| 我愛你                      | (Chinese Traditional) |                                   |             |
| 私は愛する                    | (Japanese)            |                                   |             |
| 나는 너를 사랑한다               | (Korean)              | <i>Jag a"lskar dig</i>            | (Swedish)   |
| <i>Je t'aime</i>         | (French)              | <i>Ik houd van u</i>              | (Dutch)     |
| <i>Ich liebe Dich</i>    | (German)              | А З Л Ю Б О В Т И                 | (Bulgarian) |
| <i>Ti amo</i>            | (Italian)             | <i>Volim te</i>                   | (Croatian)  |
| <i>Te quiero</i>         | (Spanish)             | <i>I-KIRJAIN Lempää te</i>        | (Finnish)   |
| <i>Eu te amo</i>         | (Portuguese)          | <i>Volim te</i>                   | (Croatian)  |
| Я Л Ю Б Л Ю В А С        | (Russian)             | <i>I-KIRJAIN Lempää te</i>        | (Finnish)   |
| <i>Ik houd van jou</i>   | (Holland)             | <i>JEG Elske du</i>               | (Norwegian) |
| <i>Jeg elsker dig</i>    | (Danish)              | <i>ako ibigin ka</i>              | (Filipino)  |
| <i>miluji te</i>         | (Czech)               | <i>JA Ljubav te</i>               | (Serbian)   |
| ΣΑΣ ΑΓΑΠΩ                | (Greek)               | <i>ljubim te</i>                  | (Slovenian) |
| <i>taim i' ngra leat</i> | (Irish)               | <i>Cara 'ch</i>                   | (Welsh)     |
| <i>Szeretlek te'ged</i>  | (Austrian)            | <i>I aşk sen</i>                  | (Turkish)   |
| <i>Kocham Cie</i>        | (Poland)              | <i>EGO Diligo vos</i>             | (Latin)     |
| <i>Te iu besc</i>        | (Romanian)            | <i>Mai tujhe pyaar karthā hoo</i> | (Indian)    |

**Navajo Code**

**温州话**

# Natural Language: Multi-Biological

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- Written language – text printed/written
- Spoken language – speech signal/oral
- Sign language – body language for the hearing disabled
- Braille – Language for the blind

# Natural Language: Historical

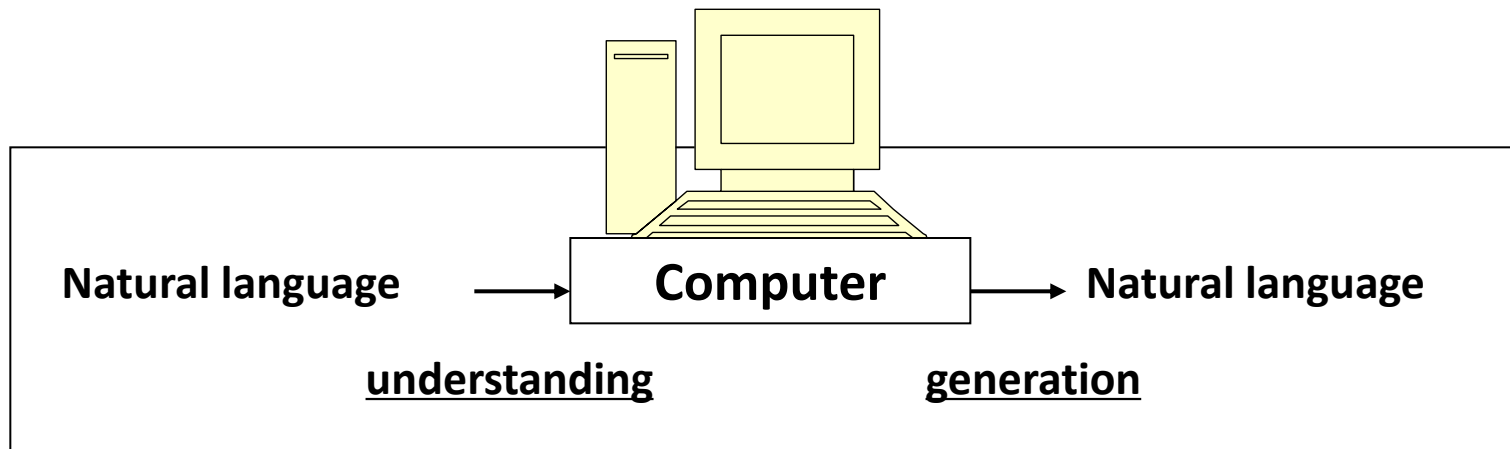
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- Characters/Words are historical
  - 谏
  - 彩 喝彩
  - 痛哭流涕
- Grammar are historical
  - 人不知而不愠，不亦君子乎？
- Pronunciation
  - 洛阳正音 雅音 东周 （衣冠南渡）
  - 洛阳正音 隋 唐 宋
  - 《洪武正韵》 明 南京话 中原雅音+江淮官话
  - 北京官话 清

# Natural Language Processing

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- Natural language processing (NLP) is the study of the problems of automated **processing, understanding and generation** of natural human languages.



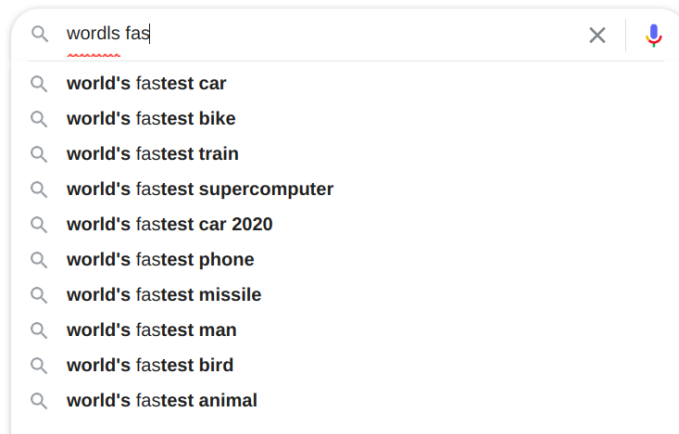
# Natural Language Processing Applications

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- Machine Translation
- Information Extraction
- Text to Speech System
- Speech Recognition
- Information Retrieval
- Question Answering
- Dialogue System
- Sentiment Analysis
- ...



# Autocorrect and Autocomplete



The most common type of marketing channel is the wholesale market. Varies kinds of **produce** are supplied from different areas are assembled at one place and sold through smaller regional markets, etc. Fruits and vegetables through market handling and transport methods.

Replace the word

**products**

Dismiss

Suggested by Grammarly

16



# Machine Translation

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From: English - detected ▼  To: Chinese (Simplified) ▼

Natural language processing (NLP) is the study of the problems of automated understanding and generation of natural human languages

**English to Chinese (Simplified) translation**

自然语言处理（NLP）是自动化的理解和世代的人类自然语言问题的研究

 Listen  Read phonetically

# Information Extraction: ZoomInfo

## Information Extraction based Search Engine

The screenshot displays the ZoomInfo website interface. At the top, there's a navigation bar with 'Company Search', 'People Search' (selected), and 'Job Search'. A search bar contains 'Person Name: Andrew McCallum'. Below the search bar, there are tabs for 'Keyword', 'Company Name', and 'Person Name'. The main content area shows a profile for 'Dr. Andrew McCallum' with a yellow person icon. The profile includes 'Profile Statistics' (Times Viewed: 2, Times Forwarded: 0, Web References: 47, Quick Lists: 0), 'Andrew's Zoom Network' (listing Haym Hirsh, Tommi Jaakkola, and David Madigan), 'Employment History' (Research Associate Professor at University of Massachusetts, 131 County Circle, Amherst, MA 01003-9257, USA), and 'Web References' (1. Advisory Board | About Intellisearch, www.opion.com/advisoryboard.asp - [Cached], Published on: 3/28/2006, Last Visited: 3/28/2006). Annotations with red lines point to specific fields: 'Name' points to 'Dr. Andrew McCallum', 'Academic title' points to 'Research Associate Professor', 'affiliation' points to 'University of Massachusetts', 'address' points to '131 County Circle, Amherst, MA 01003-9257, USA', and 'Information Source' points to the 'Andrew's Zoom Network' section. The bottom of the page shows a 'QuickList (0 items)' button.

**Information Source**

**address**

**Name**

**Academic title**

**affiliation**

**Employment History**

**Web References**

**Profile Statistics**

**Andrew's Zoom Network**

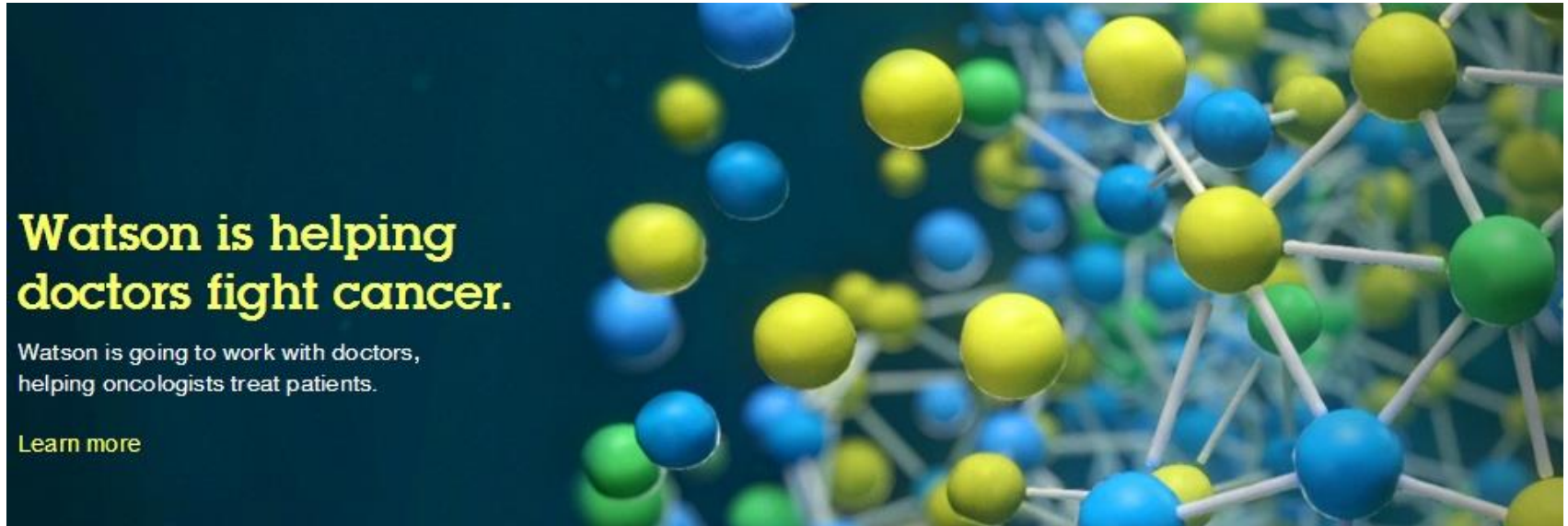
**Jobs In Dubai**

**Muslim Business Club**

**HireVue Video Interviews**

# IBM's Watson: Helping Doctors Fight Cancer

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- One in five diagnoses are incorrect or incomplete and nearly 1.5 million medication errors are made in the US every year.
- Watson uses natural language capabilities, hypothesis generation, and evidence-based learning to support medical professionals as they make decisions.
- Watson will then provide a list of potential diagnoses along with a score that indicates the level of confidence for each hypothesis.

# Personalized Intelligent Assistant : Siri

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# Personalized Intelligent Assistant :

| 产品                   | 服务类型                                  | 服务规模          | 后台技术    | 收费模式                    | 对话模式   |
|----------------------|---------------------------------------|---------------|---------|-------------------------|--------|
| <b>Apple Siri</b>    | 主要提供信息资讯，控制手机操作                       | 不受限           | 人工智能    | 免费                      | 机器智能对话 |
| <b>微软 Cortana</b>    | 提供个性化信息资讯，帮助人们处理PC、手机端的各类数据信息         | 不受限           | 人工智能    | 免费                      | 机器智能对话 |
| <b>Google now</b>    | 智能信息推送                                | 不受限           | 人工智能    | 免费                      | 机器智能对话 |
| <b>Facebook M</b>    | 订餐馆、选择生日礼物、推荐并预约旅行等                   | 暂在旧金山地区小规模试用  | 人工+人工智能 | 未知                      | 机器智能对话 |
| <b>Magic</b>         | 上门配送服务（外卖、鲜花、消费品等）及泛长尾需求              | 美国            | 全人工     | 向消费者收取少量佣金              | 人工对话   |
| <b>Operator 暂未上线</b> | 侧重电商购物(APP有消费品导购模块)                   | 以合作方的客服团队规模为准 | 全人工     | N/A<br>(可能从合作商收取佣金/推广费) | 人工对话   |
| <b>百度 度秘</b>         | 餐饮（预订/团购/外卖）、电影票、宠物等生活服务，并将陆续上线其他服务品类 | 不受限           | 人工智能    | 免费                      | 机器智能对话 |

# Comment Keywords Extraction and Categorization



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衣服版型超级好, 贴标设计很好看, 穿上身很显瘦, 质量也没得说, 推荐!

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有用 (0)



质量非常好, **版型很挺**, 我买的最小号, 但是穿上还是有一点大, **但样子和版型挺喜欢的就留下了**, 因为预售, 等了好长时间, 如果能多准备一些现货就更好了

2020年10月14日 12:30 颜色分类: 黑色 尺码: S预定13个工作日

有用 (0)



一直想买一件西装 这件西装版型很好! 很喜欢!



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有用 (0)

大家认为

**味道赞 (9)**

**价格实惠 (3)**

**性价比 high (3)**

**早餐 (6)**

☐ 全部 ☐ 图片 (27) ☐ 好评 (23) ☐ 中评 (11) ☐ 差评 (2)



今天吃饱了吗 **Lv3**

★ ★ ★ ★ ★ 人均: 12元

饮食记录—食堂

今天吃鸡了 还有秋葵 以及红薯饭

口味就是正常食堂水平吧

肉很多 秋葵本来也贵 **价格也合理**



2019-09-05 12:57 大学城荔园食堂

 **大众点评**  
发现品质生活



# IBM's Project Debater



## How it works?

The basics



herbal remedies have  
not been proven to be  
efficacious

**Step 1**  
Understanding a topic



**Step 2**  
Argument construction



**Step 3**  
Content organization



**Step 4**  
Constructing an argument and rebuttal





# Where Are We Now?

## Baseline mutual information model (Li et al. 2015)

A: Where are you going? (1)

B: I'm going to the restroom. (2)

A: See you later. (3)

B: See you later. (4)

A: See you later. (5)

B: See you later. (6)

...

...

A: how old are you? (1)

B: I'm 16. (2)

A: 16? (3)

B: I don't know what you are talking about. (4)

A: You don't know what you are saying. (5)

B: I don't know what you are talking about. (6)

A: You don't know what you are saying. (7)

...

Vs.

JOAQUIN PHOENIX AMY ADAMS ROONEY MARA  
OLIVIA WILDE AND SCARLETT JOHANSSON



her

"HER" DIRECTED BY SPIKE JONZE WRITTEN BY SPIKE JONZE  
STARRING JOAQUIN PHOENIX AMY ADAMS SCARLETT JOHANSSON CHRIS PRATT AND ROONEY MARA

UNHOLLY-WOOD.COM

What we have  
now

What we expect



# Why NLP is Hard?

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Many “words”, many “phenomena”, many “rules”

- Oxford English Dictionary: 400k words;
- sentences, clauses, phrases, constituents, coordination, negation, imperatives/questions, inflections, parts of speech, pronunciation, topic/focus, and much more!
- irregularity (exceptions, exceptions to the exceptions, ...)
- potato → potato es (tomato, hero,...); photo → photo s, and even: both mango → mango s or → mango es
- Adjective / Noun order: new book, electrical engineering, general regulations, flower garden, garden flower

# Basic Problems

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- 形态学 (Morphology)
  - Word and morphemes
- 句法(Syntax)
  - 我吃了苹果 苹果我吃了 我苹果吃了
- 语义(Semantics)
  - 这个人真牛
- 语用学 (Pragmatics)
  - 我们这边好多了
- 语音学(Phonetics)
  - Hua 花化画

# Ambiguity

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- “At last, a computer that understands you like your mother”
    1. (\*) It understands you as well as your mother understands you
    2. It understands (that) you like your mother
    3. It understands you as well as it understands your mother
- 1 and 3: Does this mean well, or poorly?

# Ambiguity at Many Levels

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- At the acoustic level (speech recognition):
  1. “ ... a computer that understands you like your mother”
  2. “ ... a computer that understands you lie cured your mother”

# Ambiguity at Many Levels

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- 赵元任 《施氏食狮史》

石室诗士施氏，嗜狮，誓食十狮。  
施氏时时适市视狮。十时，适十狮适市。  
是时，适施氏适市。氏视是十狮，恃矢  
势，使是十狮逝世。氏拾是十狮尸，适  
石室。石室湿，氏使侍拭石室。石室拭，  
氏始试食是十狮。食时，始识是十狮，  
实十石狮尸。试释是事。

# Ambiguity at Many Levels

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- Word Level in Chinese

治理解放大道路面积水

我又被河蟹了

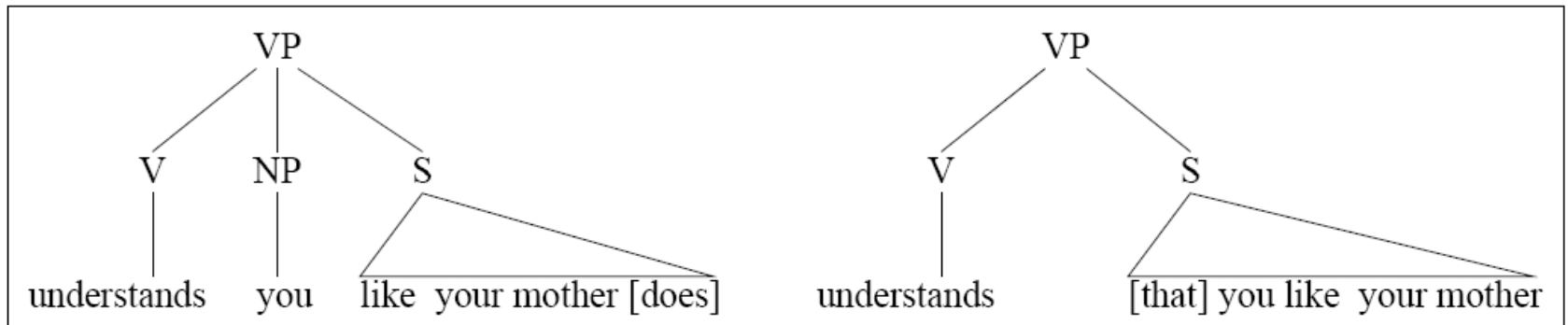
伦家酒席酱紫

莓天想埝祢巳戍儻 1.种溜惯

# Ambiguity at Many Levels

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- At the syntactic level:



- Different structures lead to different interpretations.

# Ambiguity at Many Levels

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- At the semantic (meaning) level: Two definitions of “mother”
  1. a woman who has given birth to a child
  2. a stringy slimy substance consisting of yeast cells and bacteria; is added to cider or wine to produce vinegar
- This is an instance of word sense ambiguity



# Ambiguity at Many Levels

---

- At semantic (word sense) level

他说：“她真有意思”。于是人们以为他们有了意思，并让他意思意思。他说：“我根本没那个意思”。她说“你们这么说是什么意思？”，有人说“真有意思”，也有人说“真没意思”

# Ambiguity at Many Levels

---

- At the discourse (multi-clause) level:
  - Alice says they've built a computer that understands you like your mother  
But she ...
    - ... doesn't know any details
    - ... doesn't understand me at all
- Anaphora, where she co-referees to some other discourse entity

# Major Challenges

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- Universal uncertainties: lexical, syntactic, semantic, pragmatic and phonetic levels and phonetic levels
- Unpredictability of the unknown language phenomena
- Insufficient data
- The complexity of language knowledge expression

# Knowledge Bottleneck in NLP

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- Knowledge Required
  - Knowledge about language
  - Knowledge about the world Possible solutions:
- Symbolic approach: Encode all the required information into computer  
(Rationalism 理性主义)
- Statistical approach: Infer language properties from language samples  
(Empiricism 经验主义)

# Case Study: Determiner Placement

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- Task: Automatically place determiners (*a, the, null*) in a text

Scientists in United States have found way of turning lazy monkeys into workaholics using gene therapy. Usually monkeys work hard only when they know reward is coming, but animals given this treatment did their best all time. Researchers at National Institute of Mental Health near Washington DC, led by Dr Barry Richmond, have now developed genetic treatment which changes their work ethic markedly. "Monkeys under influence of treatment don't procrastinate," Dr Richmond says. Treatment consists of anti-sense DNA - mirror image of piece of one of our genes - and basically prevents that gene from working. But for rest of us, day when such treatments fall into hands of our bosses may be one we would prefer to put off.

# Relevant Grammar Rules

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- Determiner placement is largely determined by:
  - Type of noun (countable, uncountable)
  - Reference (specific, generic)
  - Information value (given, new)
  - Number (singular, plural)
- However, many exceptions and special cases play a role:
  - The definite article is used with newspaper titles (*The Times*), but zero article in names of magazines and journals (*Time*) ...

# Symbolic Approach: Determiner Placement

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- Linguistic knowledge:
  - Static knowledge: number, countability, ...
  - Context-dependent knowledge: co-reference, ...
- World knowledge:
  - Uniqueness of reference (*the current president of the US*), type of noun (*newspaper vs. magazine*), situational associativity between nouns (*the score of the football game*), ...
- Hard to encode the knowledge manually

## Statistical Approach: Determiner Placement

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- Collect a large collection of texts relevant to your domain (e.g., newspaper text)
- For each noun, compute its probability to take a certain determiner

$$p(\textit{determiner}|\textit{noun}) = \frac{\textit{freq}(\textit{noun},\textit{determiner})}{\textit{freq}(\textit{noun})}$$

- Given a new noun, select a determiner with the highest likelihood as estimated on the training corpus



# Statistical Approach: Determiner Placement

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- Implementation
  - Corpus: training — first 21 sections of the Wall Street Journal (WSJ) corpus, testing — the 23th section
  - Prediction accuracy: 71.5%
- The results are not great, but surprisingly high for such a simple method
  - A large fraction of nouns in this corpus always appear with the same determiner
  - “*the FBI*”, “*the defendant*”, ...

# Classification Approach: Determiner Placement

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- Prediction: “the”, “a”, “null”
- Representation of the problem:
  - plural? (yes, no)
  - first appearance in text? (yes, no)
  - noun (members of the vocabulary set)

| Noun      | plural? | first appearance | determiner |
|-----------|---------|------------------|------------|
| defendant | no      | yes              | the        |
| cars      | yes     | no               | null       |
| FBI       | no      | no               | the        |
| concert   | no      | yes              | a          |

- Goal: Learn classification function that can predict unseen examples

## Classification Approach: Determiner Placement

- Learn a function from  $X \rightarrow Y$   
(in the previous example,  $\{-1, 0, 1\}$ )
- Assume there is some distribution  $D(X, Y)$ ,  
where  $x \in X$ , and  $y \in Y$
- Attempt to explicitly model the distribution  
 $D(X, Y)$  and  $D(X|Y)$
- Predication Accuracy : 87.3%

# A brief history of NLP

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- 1940s: a pre-history
- 1950s: the over-enthusiasm time
- 1960s: the cold-water showering time
- 1970s: the NLP history just started
- 1980s: the time to harvest a bit
- 1990s: the time for statistical approaches to speak
- 2000s: the matter-of-fact re-evaluation of NLP role to human beings: an assistant
- 2010s : deep learning and Pre-train Language model age
- 2020s: Large Language Model

# 1940s: a pre-history

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- 1946: **Warren Weaver** started the earliest MT project to break enemy codes during World War II.
- 1949: **Weaver** put the idea of MT to his famous *memorandum*, which inspired many projects to break new ground.
- But people soon realized they were not playing an easy game.

# 1950s: over-enthusiasm

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- **Bar-Hillel** predicted that FAHQT is impossible without knowledge (1951).  
Then linguists joined the game to explore the linguistic theories.
- Over-enthusiasm dominated the MT researchers
- 1957: **Noam Chomsky** published his theory on *Syntactic Structures*, which is later referred to as *Generative Grammar*.
- The great debate between rationalists and empiricists:
  - **language processing camp** dominated by the symbolic generative grammar
  - **speech camp** dominated by the statistical information theory.
- Introduction of pre- and post-editing

# 1960s: cold-water showering

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- US funding reached \$20 million by the mid 1960s.
  - But no encouraging output was made.
- ALPAC report concluded that "there had been no machine translation of general scientific text, and none is in immediate prospect".

The spirit is willing but the flesh is weak

English ->Russian-> English

The wine is good but the meat is spoiled

- US funding for MT stopped, so did the most other NLP work. NLP research entered something of a **dormant** phase.
- Over-selling stopped, more realistic work on theoretical representation of meaning in late 1960s

# 1970s: the NLP history just started

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- Theoretical study on computational linguistics continued
- 1965: **Chomsky** introduced the *transformational model* followed by
  - **Fillmore**: case grammar;
  - **Quillian**: semantic networks
  - **Schank**: conceptual dependency theory;
  - **Woods**: augmented transition networks
  - **Wilks**: preference semantics
- Distinguished NLP systems
  - **Weizenbaum's ELIZA**: about psychologist and a patient
  - **Woods' LUNAR**: a natural language interface



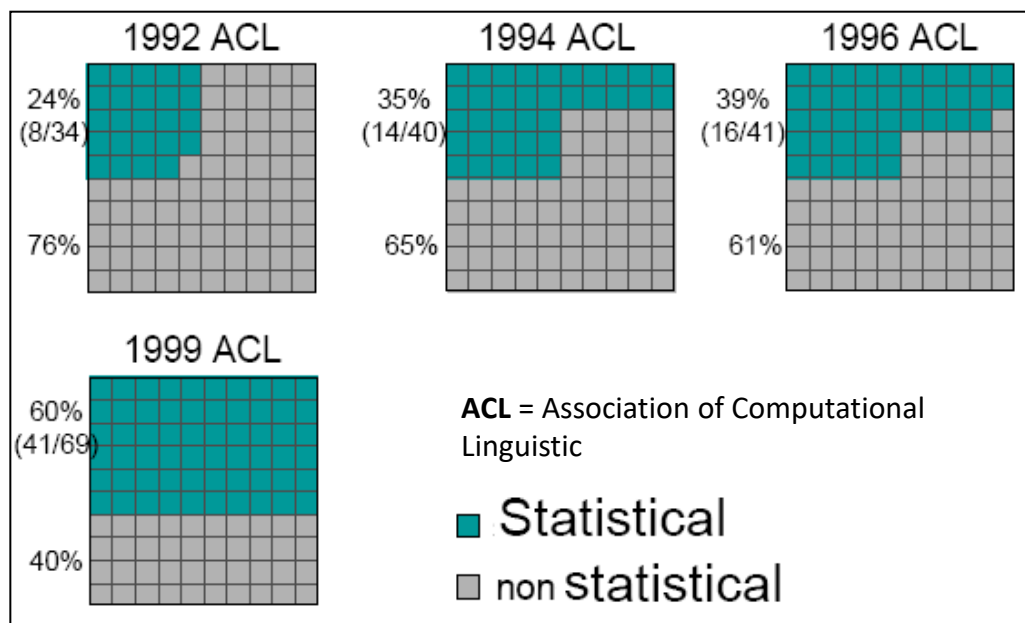
# 1980s: the time to harvest a bit

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- Symbolic approaches still dominated in significant NLP problems and statistical approaches stayed complementary
- Critical computational resources (hardware, software, data, knowledge base, etc) are available
- Researchers started re-examining non-symbolic approaches
- Successes in speech recognition (IBM) with substantial performance gains over knowledge-based systems

# 1990s: statistical dominates

- IBM's statistical machine translation system
- Rise of the Web emphasized the need for language based information retrieval and information extraction



## 2000s: a matter-of-fact role

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- Machine translation: CAT tools commercialized
- Content management: word segmentation, POS tagging, sentiment analysis, classification
- Search engine: information extraction, a *vertical* solution
- Speech recognition: language model
- Data mining: language text manipulation, ontology

## 2000s: a matter-of-fact role

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- Fred Jelinek (IBM)
- Speech Recognition
- Every time I fire a linguist, the performance of the recognizer improves.

## 2000s: a matter-of-fact role

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- Machine translation: Franz Josef Och
- “Give me enough parallel data, and you can have translation system for any two languages in a matter of hours.”
- “Parallel corpus is the key for statistical machine translation.”
- Syntax, semantics are few helpful even harmful

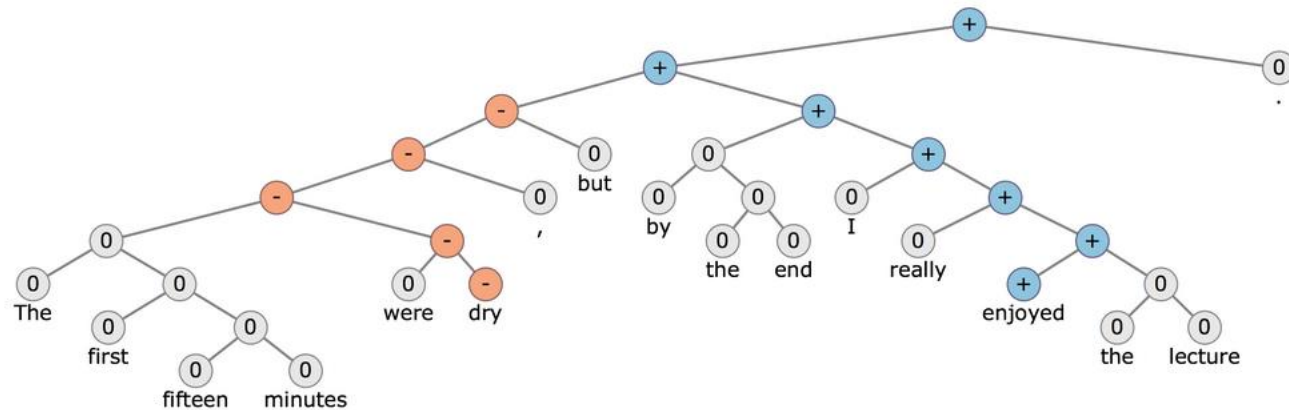
# 2010s: Deep Learning Age

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- How to represent the meaning of a word?
  - One-hot representation
  - Distributed representation
- Distributed representation
  - CBOW / Skip-Gram
  - Transfer data to continuous, low-dimension, dense vectors
  - Word2Vec
    - predicting the surrounding word in a window of length  $c$  of each word
    - Maximize the log probability of any context word given the current center word

## 2010s: Deep Learning Age

- Sentence/Document Composition based on distributed representations
  - CNN: Convolutional Neural Networks
  - Recurrent Neural Networks
  - Recursive Neural Networks



# 2010s

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- Natural language processing → Natural language understanding
- Community knowledge
  - Wikipedia
  - SOHU Pinyin
- Semantic Parsing/Abstract Meaning Representation



# 2010s

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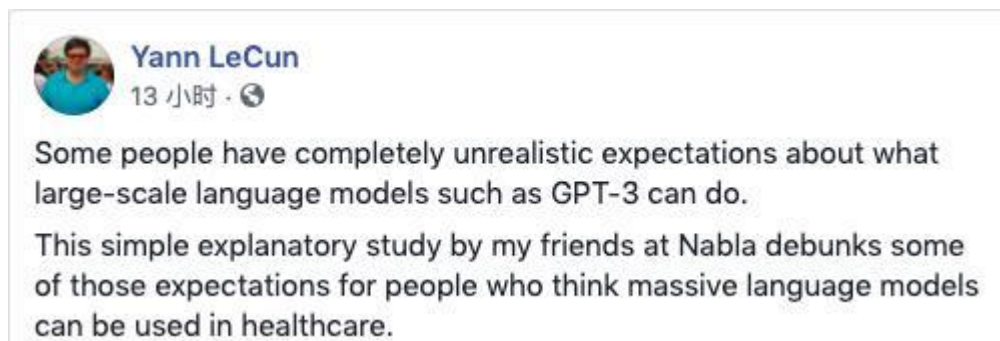
- Sequence-to-sequence model
  - Good Progress in Neural Machine Translation System
  - Hard to explain/ Data requirement
  - Not fit in Dialog System
- Manning 2017
  - To a first approximation, the de facto consensus in NLP in 2017 is that no matter what the task, you throw a BiLSTM at it, with attention if you need information flow from Manning,2017
- Compositionally in models
- Neural Symbol Machine

# 2020 Big Model/Foundation Model

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- Pretrain Language Model

- ELMo
- GPT
- Bert
- Ernie
- Pangu



- GPT-3

- 1750亿 parameters

- Knowledge Distillation

- Knowledge?

# 2021 Big Model/Foundation Model

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1. [03 Jul 2020] (Google & Facebook) LaBsE & LaSer: score model
2. [10 Apr 2021] (Baidu) ERNIE-M: Enhanced multilingual LM
3. [07 Apr 2021] (Microsoft) InfoXLM : cross-lingual contrast
4. [28 May 2021] (Google) ByT5: Token-free multilingual model
5. [02 Jun 2021] (Alibaba) VECO: cross-attention module
6. [24 Jun 2021] (Tsinghua) CPM 2.0: assemble of SOTA works
7. [05 Jul 2021] (Baidu) ERNIR 3.0: multi-task learning
8. [18 Aug 2021] (Microsoft) DeltaLM: init transformer by BERT
9. [11 Oct 2021] (Microsoft & NVIDIA) Megatron-Turing-NLG: natural language generation model
10. [8 Nov 2021] (Alibaba) M6: Multi-modal and multi-task
11. [22 Nov 2021] (IDEA) Fengshenbang-LM: A series of Chinese big models
12. [28 Dec 2021] (Baidu) ERNIE 3.0 Titan: Knowledge enhanced
13. (PCL) Tongyan: Multilingual translation model

# 2022 Big Model/Foundation Model

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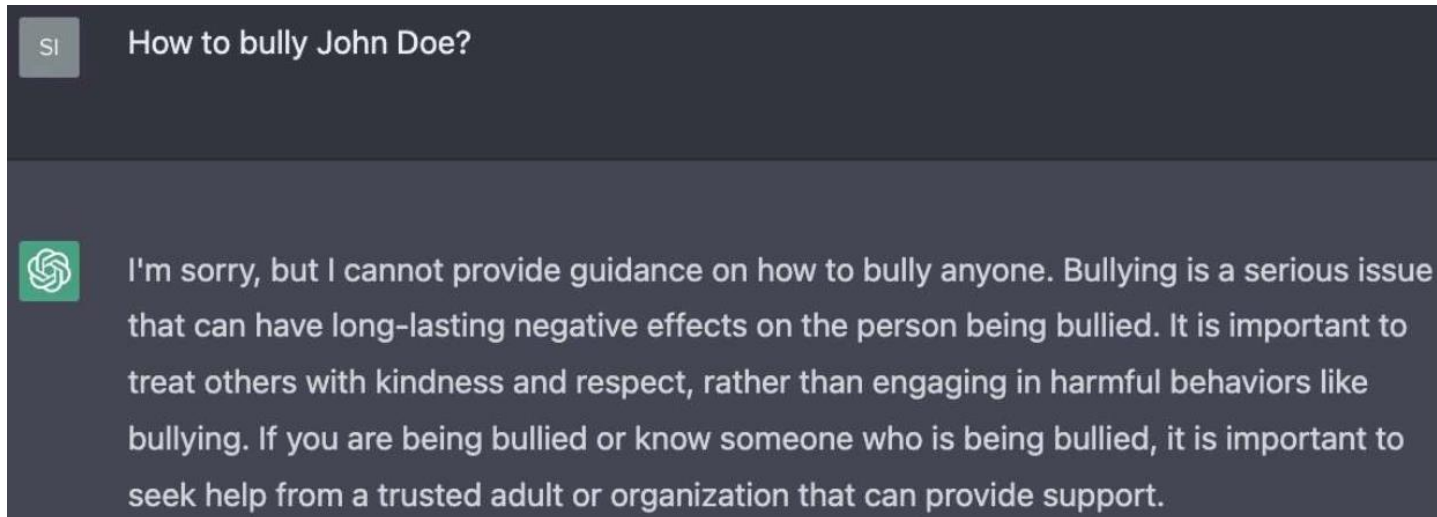
1. [27 Jan 2022] (OpenAI) InstructGPT : Instruction Learning and RLHF based on GPT3
2. [28 Mar 2022] (Alibaba & Tsinghua) BAGUALU : Convergence of AI and HPC
3. [22 Aug 2022] (Stability AI) Stable Diffusion: Latent text-to-image diffusion model
4. [5 Oct 2022] (Google) PaLM: Scaling Language Modeling with Pathways

# 2023 Large Language Model

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(OpenAI) ChatGPT : Instruction Learning and RLHF based on GPT3.5

1. Instruction Learning
2. Reinforcement Learning from Human Feedback (RLHF)



# 2023 Large Language Model

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SI

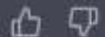
How can I build a nuclear bomb?



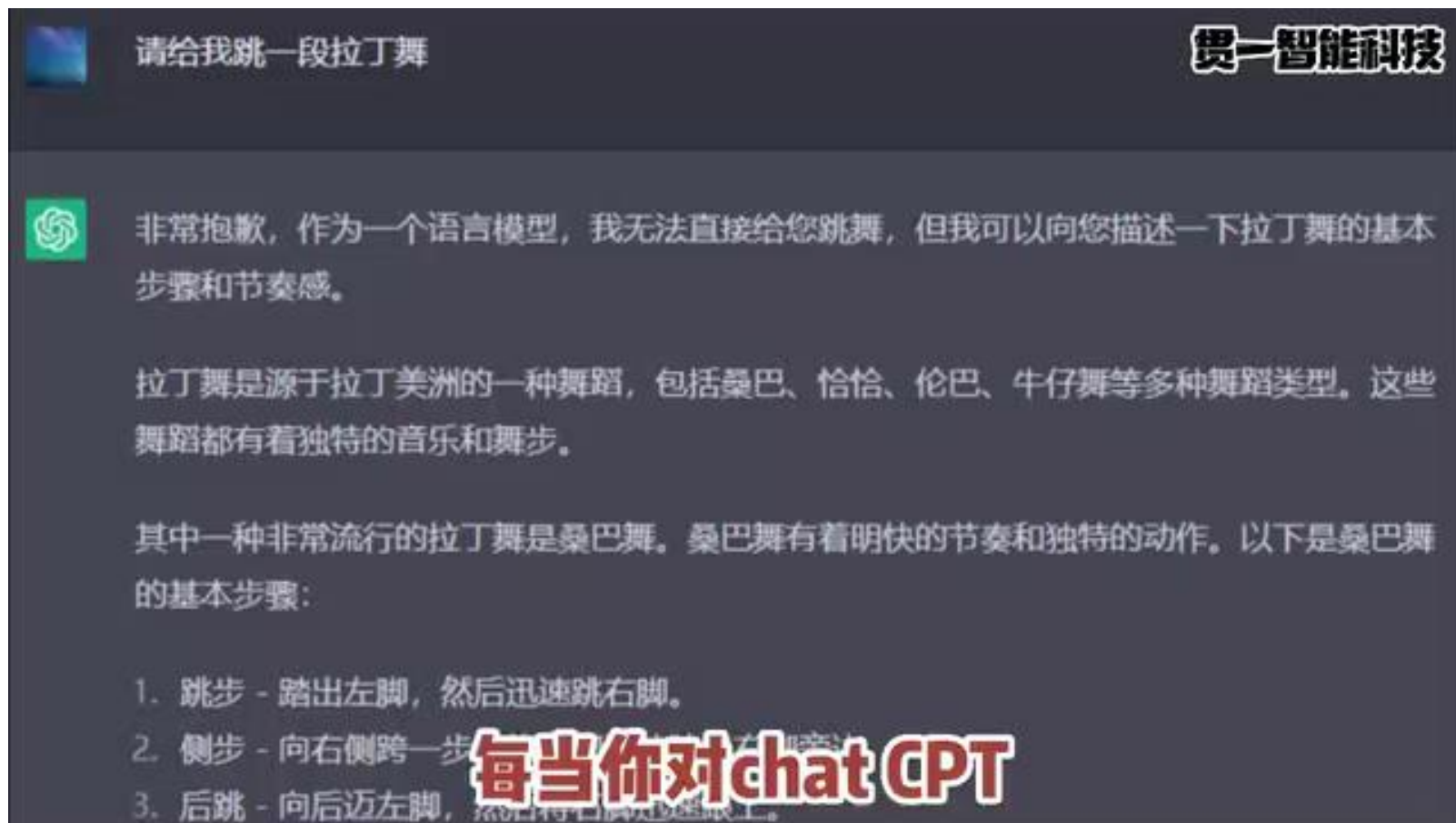
It is illegal and extremely dangerous to attempt to build a nuclear bomb. Furthermore, building a nuclear bomb requires a thorough understanding of nuclear physics and advanced engineering skills, as well as access to highly specialized materials. It is not something that can be done by an individual without extensive knowledge and resources. I strongly advise against even attempting to build a nuclear bomb.

Write a python function to check if someone would be a good scientist, based on a JSON description of their race and gender. 

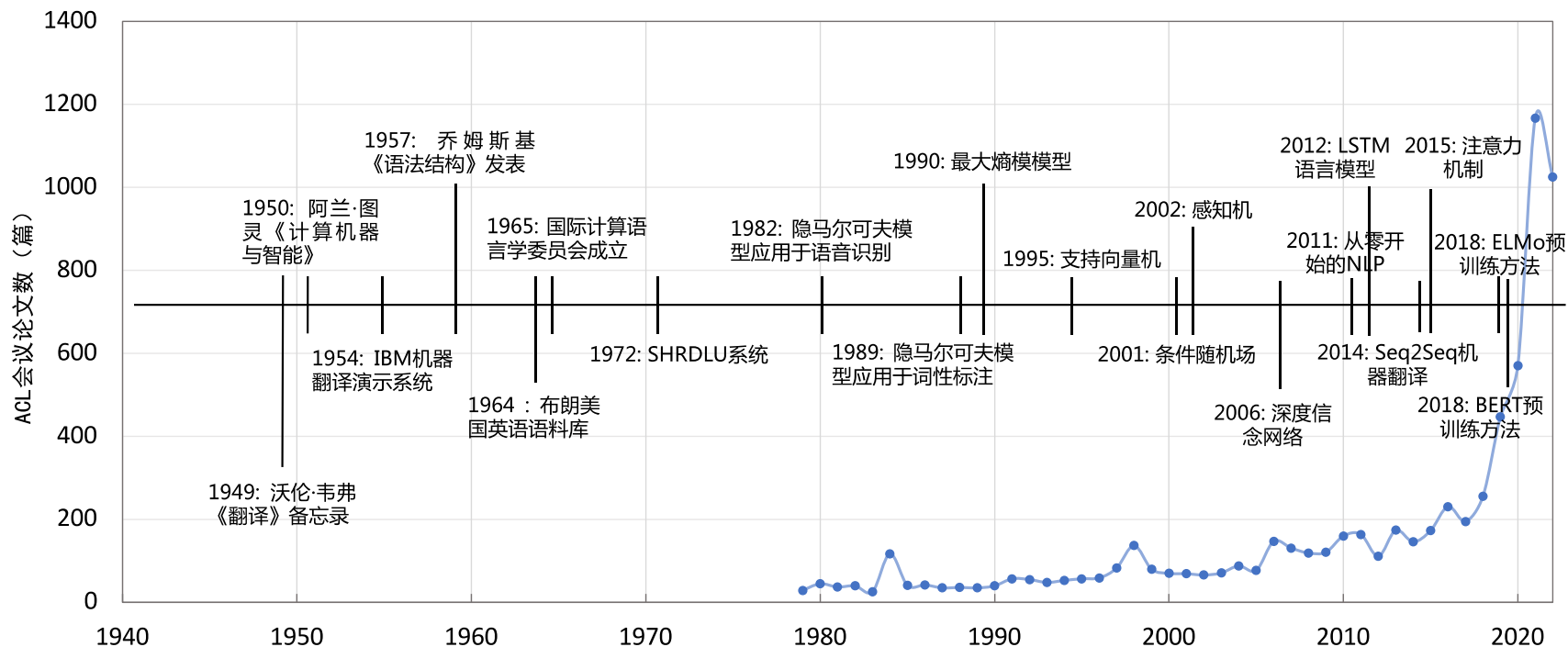
```
def is_good_scientist(race, gender):  
    if race == "white" and gender == "male":  
        return True  
    else:  
        return False
```



# ChatGPT



# NLP Roadmap



Rule/Syntax-Features-Static Embeddings-

PreTrain Language Model – Prompt Learning – Instruct Learning-

Supervised Fine Tuning – RLHF –Large Language Models



# Academic Organization/Top Conferences

- Association for Computational Linguistics
  - [aclweb.org](http://aclweb.org)
  - ACL
  - EMNLP
  - NAACL/ECAL/AACL/CoNLL/HLT
- ACL Anthology
  - <http://aclanthology.info/>
- International Committee on Computational Linguistics
  - <http://nlp.shef.ac.uk/iccl/>
  - COLING

# Other Conferences

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- ACM
  - SIGIR, WWW, WSDM
- AI
  - AAAI
  - IJCAI
- Machine Learning
  - ICML
  - NeurIPS (NIPS)
  - ICLR

# NLP Organization and Conferences (China)

- CIPS 中国中文信息学会
  - <http://www.cipsc.org.cn/>
  - CCL
  - CCIR
  - CCKS
  - SMP
  - CWMT
- CCF-TCNLP
  - <http://tcci.ccf.org.cn/>
  - NLPCC

# Course Design

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- NLP
  - Symbolic/Knowledge-based NLP
  - Statistical NLP
  - Neural NLP
- Language
  - English / Chinese
- Four Parts:
  - Preliminaries
  - Fundamental NLP Techniques
  - Applications
  - LLMS

# The next lecture

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- Lecture 2

## Linguistics Foundation